INDUSTRY PROBLEM SOLVING TASK

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Course : Computer Vision

1. Autonomous Vehicle Vision System Selection

Problem Statement

An automobile company is developing an autonomous vehicle detection system. The system must achieve an accuracy of at least 90%, have a latency of 70 ms or less, and operate using moderate computational resources.

Comparison of Available Methods

Method	Accuracy	Latency	Computation	Meets Accuracy?	Meets Latency?	Meets Computation?
Viola-Jones	82%	30 ms	Low	No	Yes	Yes
YOLO	93%	60 ms	Moderate	Yes	Yes	Yes
Deep Learning	97%	90 ms	High	Yes	No	No

The given performance values are provided in the assessment.

Logical Evaluation

Viola-Jones:

Viola-Jones has a low computational requirement and a latency of only 30 ms. However, its accuracy is only 82%, which is below the required 90%. Therefore, it fails the accuracy constraint.

YOLO:

YOLO provides 93% accuracy, which is above the required 90%. Its latency is 60 ms, which is below the maximum acceptable latency of 70 ms. It also requires moderate computation, matching the available hardware resources. Therefore, YOLO satisfies all three constraints.

Deep Learning:

Deep Learning provides the highest accuracy of 97%. However, its latency is 90 ms, which exceeds the maximum acceptable 70 ms. It also requires high computational resources, while the vehicle supports only moderate computation. Therefore, it does not satisfy all constraints.

Decision

YOLO is the most suitable method for the autonomous vehicle vision system.

The logical condition can be represented as:

Accuracy $\geq$ 90% AND Latency $\leq$ 70 ms AND Computation $\leq$ Moderate

For YOLO:

93% $\geq$ 90% AND 60 ms $\leq$ 70 ms AND Moderate = Moderate

Therefore, all conditions are satisfied.

Conclusion

YOLO is selected because it provides a good balance between accuracy, latency, and computational requirements. Although Deep Learning has higher accuracy, its latency and computational requirements make it unsuitable. Viola-Jones is computationally efficient but fails to achieve the required accuracy. Hence, YOLO is the best choice for the autonomous vehicle system.

2. Smart Factory Product Inspection Selection

Problem Statement

A manufacturing company wants to deploy a real-time computer vision system for detecting defective products on a production line. The system must have an accuracy of at least 95%, processing time of 50 ms or less, and must operate with moderate computational resources.

Comparison of Available Methods

Method	Accuracy	Processing Time	Computation	Meets Accuracy?	Meets Time?	Meets Computation?
Traditional Image Analysis	91%	25 ms	Low	No	Yes	Yes
YOLO	96%	45 ms	Moderate	Yes	Yes	Yes
Deep Learning	98%	75 ms	High	Yes	No	No

The performance values above are given in the assessment.

Logical Evaluation

Traditional Image Analysis:

Traditional Image Analysis has a low processing time of 25 ms and requires low computational resources. However, its accuracy is only 91%, which is below the required 95%. Therefore, it cannot reliably satisfy the defect-detection requirement.

YOLO:

YOLO provides 96% accuracy, which is greater than the required 95%. Its processing time is 45 ms, which is within the maximum limit of 50 ms. It also requires moderate computation, which matches the available hardware. Therefore, YOLO satisfies all the given conditions.

Deep Learning:

Deep Learning achieves the highest accuracy of 98%. However, its processing time is 75 ms, exceeding the maximum permitted 50 ms. It also requires high computational resources, while the available hardware supports only moderate computation. Therefore, it is not suitable for this application.

Decision

YOLO is the most suitable method for the smart factory product inspection system.

The logical condition is:

Accuracy ≥ 95% AND Processing Time ≤ 50 ms AND Computation ≤ Moderate

For YOLO:

96% ≥ 95% AND 45 ms ≤ 50 ms AND Moderate = Moderate

Therefore, all conditions are satisfied.

Conclusion

YOLO is the best choice for real-time product inspection because it satisfies the required accuracy, processing-time, and computational constraints simultaneously. Traditional Image Analysis is faster but does not provide sufficient accuracy, while Deep Learning provides better accuracy but exceeds the processing-time and computational limits. Hence, YOLO is the most appropriate solution for the smart factory application**.**